DIGITOMICS

Student Digital Behaviour Analysis

100K Students | 50+ Countries | 48 Features

Data Analytics Project | Academic Year 2024–25

Group 6

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The Digital Challenge

→ *How does digital behaviour impact student performance & wellbeing?*

5+ Hours Online Daily

Average student screen time exceeds 5 hrs/day

Rising Mental Health Issues

Anxiety & depression rates climbing among students

Academic Decline

Attention span dropping — avg. 53.68 min focus only

Impulse Spending

Digital ads driving unplanned financial decisions

KEY INDICATORS

19.02 Avg Brain Rot Index	27.99 Avg Academic Risk Score
12.54% Cyberbullying Exposed	14.08 Digital Addiction Score

Dataset at a Glance

100K

Students

Across 50+ countries

48

Features

Behavioural & demographic

15-25

Age Range

Years of age

50+

Countries

Global coverage

FEATURE CATEGORIES

- Mental Health
- Academic
- Digital Usage
- Financial
- Demographic
- Social Media

Source: Digitomics Survey Dataset 2024 • kaggle.com/datasets/nitikachandel95/student-social-media-impact-dataset/data

Data Processing Pipeline

01

Data Ingestion

100K raw records imported
CSV/JSON formats

02

Cleaning

Remove duplicates
Standardize formats

03

Imputation

Handle missing values
Median/mode strategy

04

Outlier Handling

IQR-based detection
Cap & floor method

05

Feature Engineering

Brain Rot Index
Risk scores created

RESULT: 0 null values • Clean, analysis-ready dataset • All features validated • Ready for modelling

Core Measurement Pillars

Brain Rot Index

19.02

Avg. across all students

Academic Risk

27.99

Risk score (0–100)

Digital Addiction

14.08

Addiction score (0–30)

Wellbeing Index

57.28

Composite wellbeing

Financial Risk

41.61

Avg. risk score

Late Night Usage

12%

Always online at night

What the Data Tells Us

2×

Late-Night → Anxiety

Students with always late-night usage show 2× higher anxiety scores vs those who never use devices at night

#1

Less Sleep → High Stress

Sleep hours show the strongest negative correlation with stress levels across all variables measured

21%

Social Media → Productivity Drop

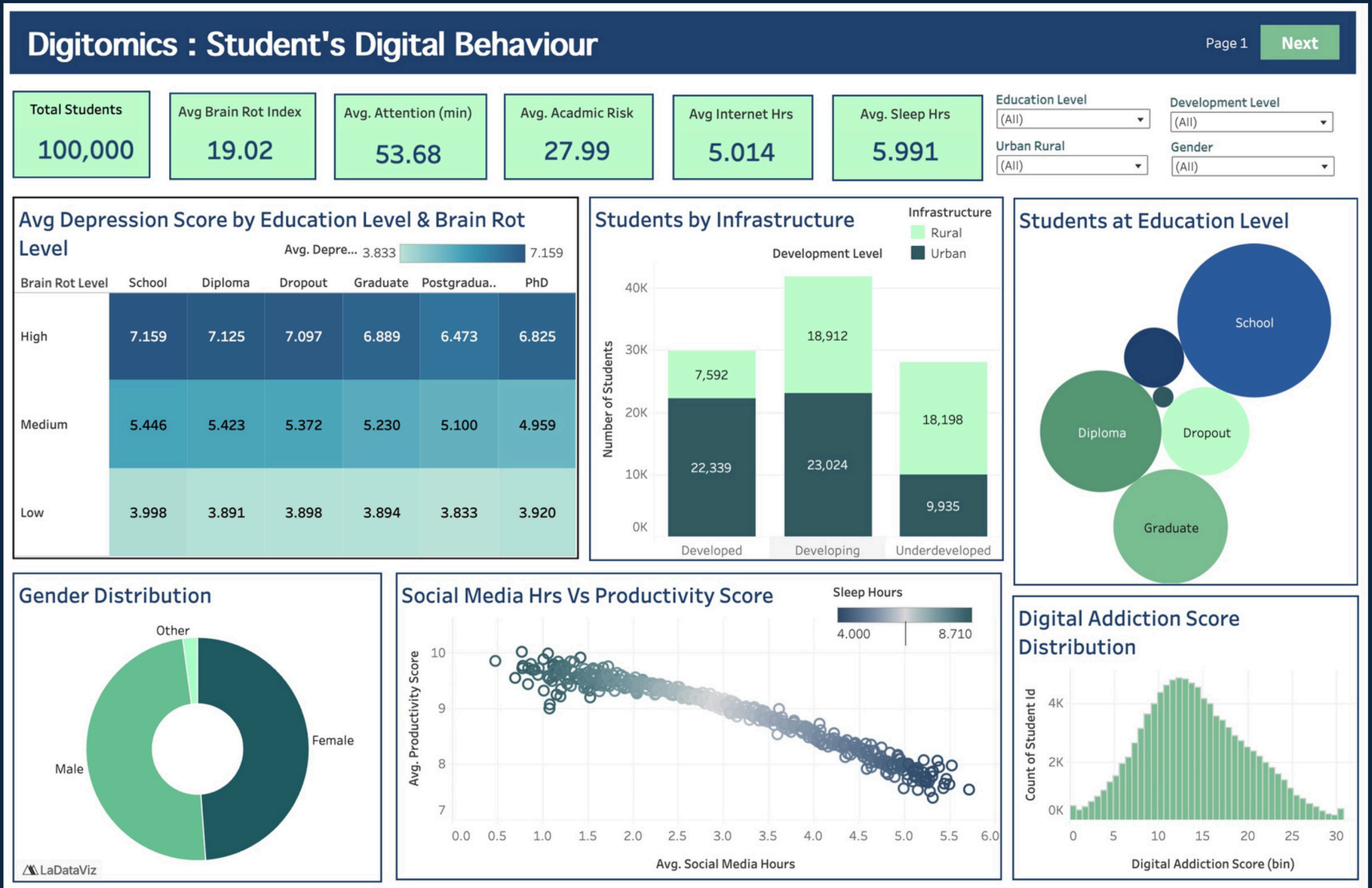
Every 2-3 hour increase of social media usage correlates with a 21% decrease in average productivity score

2×

Brain Rot Doubles Depression

Students with High brain rot index score depression levels nearly double those with Low brain rot scores (7.1 vs 3.9)

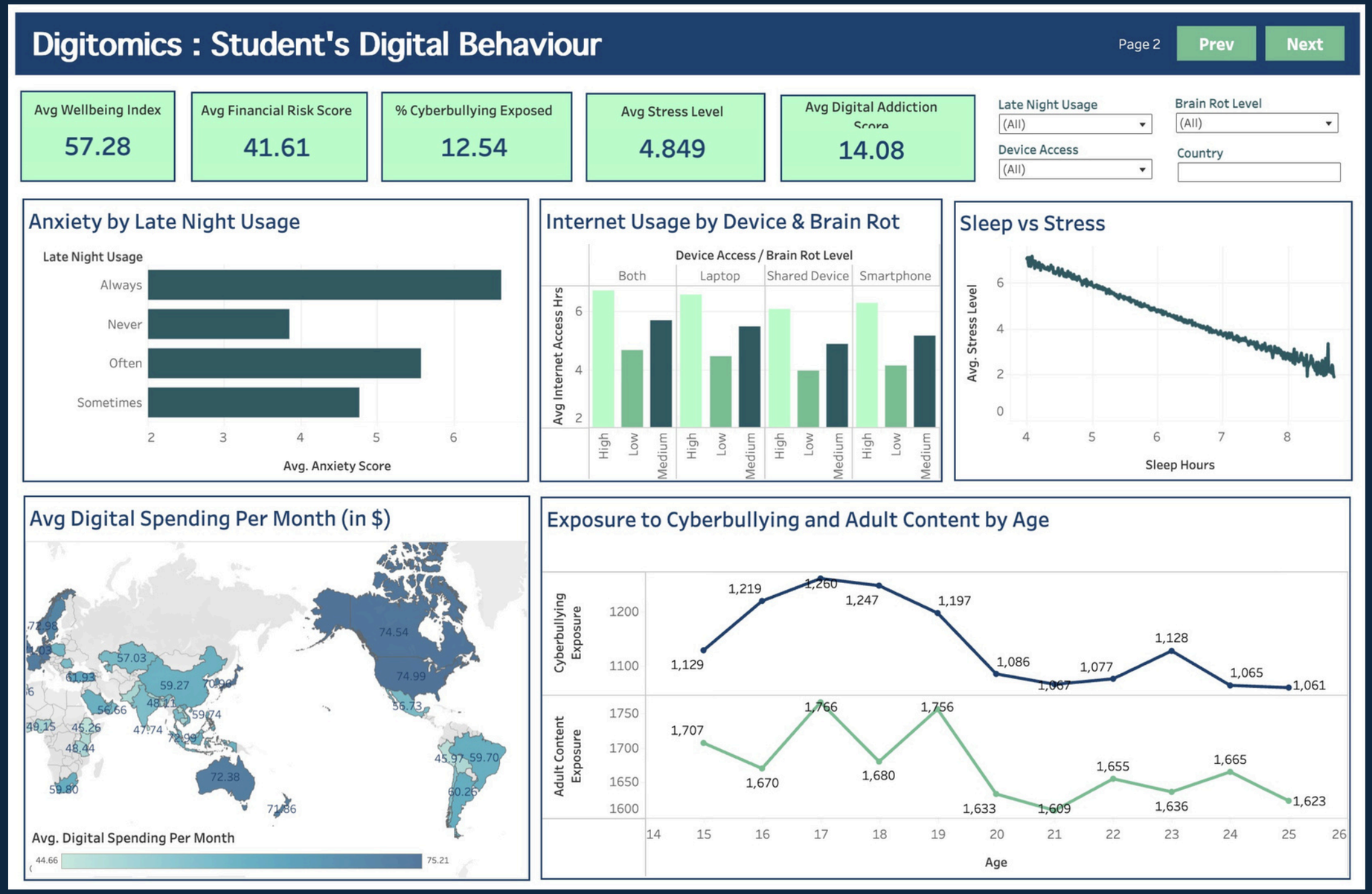
Key Insights



Brain Rot vs Depression: High BRI students score 7.1 avg depression — 82% higher than Low BRI peers (3.9). Heatmap confirms strong positive correlation ($r \approx 0.74$).

Productivity Drop: As we increase social media hrs productivity decrease. Line chart shows steepest decline between 3–5 hrs/day — the critical overuse threshold for academic risk.

Key Insights

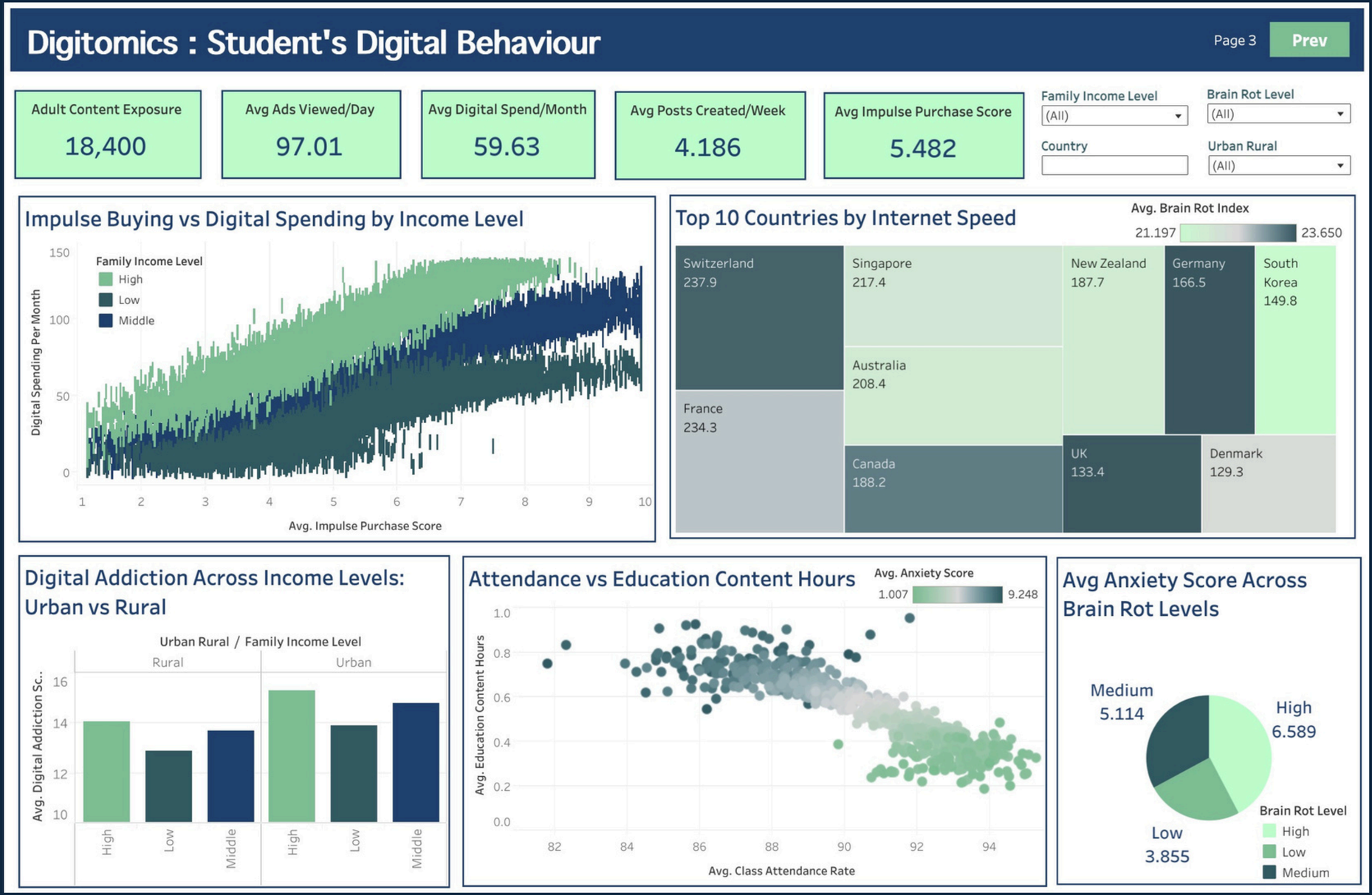


Sleep as a Stress Buffer: Students sleeping 8+ hrs report avg stress score of 3.2 vs 7.8 for those sleeping <5 hrs. Line plot confirms sleep hours as the #1 negative predictor of stress (strongest negative correlation across all 48 features).

Anxiety by Usage Timing: "Always late-night" users avg anxiety score 6.3 vs 3.5 for "Never" — a ~2× spike.

Ages 16–18 are the peak window for cyberbullying exposure and adult content exposure. With 1260 students reported for cyberbullying and 1766 students for adult content exposure

Key Insights



Digital spending rises steadily with impulse scores across all income groups. High-income users increase from ~$40 to ~$150, middle-income from ~$20 to ~$130, and low-income from ~$10 to ~$80, showing a consistent but income-driven trend.

Addiction by Segment: High-income urban students show peak digital addiction score (15.6 avg). Grouped bar chart reveals urban-rural gap widens with income — suggesting access + affluence compound addictive usage patterns.

There is a negative relationship between attendance and education content hours—students with higher attendance (~92–95%) tend to spend less time (~0.2–0.4 hrs) on additional educational content.

Recommendations

Device Curfew

Enforce no-device policy after 10 PM in student residences. Implement app-based screen time locks. Use reward-based systems for students maintaining healthy usage

Sleep & Wellness Programs

+1 hour of sleep correlates with significantly lower stress. Structured sleep hygiene workshops. Weekly sleep tracking challenges using wearables

Cyberbullying Prevention

Mandatory digital safety curriculum. Use AI-based monitoring for harmful keywords in student forums/chats. Peer support networks for exposed students (12.5%).

Financial Literacy

Digital spending awareness programs. Teach impulse control and ad-recognition skills. Show students monthly digital spending dashboards

Expected Impact

30-40%

Less late-night usage

+1 hr

Sleep → Lower stress levels

21%

Productivity loss reduction

15%

Reduction in impulse spending

⚠ Limitations: Self-reported survey data; cross-sectional design limits causal inference; sampling bias possible across regions.

THANK YOU

Empowering Students Through Data-Driven Insights

Questions & Discussion

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Students Analysed

50+

Countries Represented

48

Features Tracked